

Robust Keystroke Classification under Adversarial Conditions: A Multi-Model Evaluation with Secure Audit Framework

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I. INTRODUCTION

Acoustic side-channel attacks are based on the unintended physical noises that are generated by physical devices when they are in normal operation. With keyboards, the act of individual keys being pressed results in specific acoustic signatures which can be recorded and examined in order to determine the input typed without access to the target system itself. This renders acoustic keystroke inference a major issue of concern in terms of security as well as privacy and this is more so when the microphones are easily accessible. Experiments that have been conducted before have revealed that machine learning models can be used to classify keystroke sounds in controlled conditions, which proves that the threat is not purely theoretical.

The development of research in the field has changed the handcrafted spectral feature and traditional classifiers to deeper learning and hybrid architectures. The initial methods used feature representations like MFCCs with statistical models whereas the latest efforts have used convolutional and attention networks to learn discriminative time-frequency patterns directly using audio data. Whereas these methods have enhanced the performance of classifiers, most of the studies have placed more emphasis on accuracy when predicting in controlled settings with less emphasis on the robustness of the classifiers when faced with adversarial settings.

Security wise, this loophole is important. A model that works well on clean data may not always work on miscellaneous instances where minor, well-crafted perturbations can always lead to misclassification. This indicates the drawback of accuracy as an independent measure. In practice, models also need to be tested on their robustness to input perturbations, both to environmental noise and (adversarial) intentional perturbation. The present project thus aims at studying model

behaviour when adversarial perturbation is applied, to have a better understanding of the security limits of the acoustic keystroke inference systems.

To help solve this, the project constructs a comparative evaluation framework on a modular processing pipeline that is comprised of a sensor node, a preprocessing and feature transformation stage, an adversarial proxy, a machine learning inference engine, and a blockchain-based audit element. The system works with actual audio of recorded keystroke, and converts the audio to a feature representation, in the form of MFCCs and log-Mel spectrograms, according to the specifications of the model. Various model architecture versions are taken into consideration such as conventional machine learning models, sequential deep learning models, convolutional networks, and hybrid CoAtNet-based models.

The limitation of the experimental design is that all the models are introduced using single-keystroke samples instead of typing sequences. This guarantees that the input is represented uniformly with different types of a model and fine-grained analysis of classification behaviour is made possible. Although sequence-level modelling is a direction of importance, this approach enables a controlled comparison of the effect of individual keystroke responses on adversarial perturbation.

Adversarial assessment is conducted on several attack models, which are FGSM, BIM, PGD, and DFT-based perturbations. The gradient-based attacks are only applied to differentiable models, and other non-differentiable models, including Support Vector Machines and Random Forests, are tested with other perturbation methods. This will make sure that every model is tested with the methods that are suitable to their structure but that they are comparable among all the system.

It is a twofold contribution of work. To begin with, it offers an integrated experimental system of assessing acoustic keystroke classifiers under controlled adversarial situations.

Second, it provides a comparative study of robustness between the various model architectures which alters the emphasis of raw accuracy to a more general view of reliability, resilience and pragmatic vulnerability of acoustic side-channel inference.

II. BACKGROUND AND RELATED WORK

Acoustic side-channel attack uses the unintended sound that is emitted when a device interacts with its environment to deduce some sensitive data. Regarding keyboards, a unique acoustic signature is produced with every keystroke, which may be recorded and processed to reconstruct typed data. Such attacks have been shown in previous work to be possible provided that the conditions are controlled and that supervised learning models can attain meaningful accuracy in keystroke classification.[1],[2],[3],[4]

Initial methods were based on handcrafted feature extraction methods and in particular Mel-Frequency Cepstral Coefficients (MFCCs) alongside classical machine learning algorithms, like Support Vector Machines and Random Forests. Although useful in restricted conditions, such approaches are restricted due to their reliance on engineered phenomenon and lack of recognition of time variability of keystroke audio.[5],[6]

Current studies have moved to deeper learning designs that process higher-level representations of audio signals. Recurrent networks, including Long Short-Term Memory (LSTM) networks, learn to utilize the dependencies of time within sequences of keystroke to provide a better classification[14]. Simultaneously, convolutional models based on Log-Mel spectrograms allow extracting local time-frequency patterns without engineering features manually[15],[16]. This capability is further extended to hybrid architectures, such as CoAtNet and attention-based variants [11],[17], which can model local and global relationships between the input[10],[17].

Despite these developments, the current literature mostly reports the assessment of models in idealised settings, commonly in controlled environments, constant-record settings, and shallow noise variation. Consequently, reported accuracy is not always a reflection of the reliability in the real world. More to the point, the strength of keystroke inference models in adversarial conditions has not been investigated properly yet.

FGSM, BIM, and PGD are examples of adversarial attacks that have been demonstrated in wider fields of audio and machine learning to trigger false classification by small, structured attacks[12],[13]. Nevertheless, their systematic use to the acoustic keystroke inference is little. Moreover, this is because traditional machine learning models do not accommodate gradient-based attack, and therefore, do not provide similarity in robustness evaluation of different model types.

This disparity emphasizes the fact that there is no single framework that can be applied to assess performance and resilience in adversarial conditions. To do this, it is not only necessary to compare different model architectures but also expose them to controlled adversarial stress and study their behaviour in realistic environments.

The work contributes to the current literature by proposing a multi-model evaluation system that combines adversarial testing and secure auditing to have a comprehensive definition of robustness in the acoustic keystroke classification systems.

III. SYSTEM ARCHITECTURE

The system is constructed so that it can creating, processing, and testing acoustic keystroke information in a controlled manner in a pipeline. The design encompasses the entire process that includes creation of input data, modification of the data and ultimately usage in machine learning models instead of classifying it as a separate task. This facilitates the study of the behaviour of various models in a predictable and repeatable manner.

The system is divided into five major components as follows, a sensor node, a text-to-audio stage, an adversarial proxy, a machine learning engine, and an audit layer. The roles of each part in the pipeline are different and therefore the system is very organised and easy to test and comprehend each stage individually.

A. Sensor Node

The sensor node feeds the system with the input or with the keystroke sequences that are to be tested. Such inputs may be plain text, structured data such as code or more haphazard sequences such as passwords. By regulating what is entered at this point, the system ensures that everything is in tune and in case of re-doing the same tests, it becomes easier.

B. Text-to-Audio(TTA) Converter

The text to audio converter transforms the keystroke codes to audio. It does not use real typing sounds but is able to produce audio in a controlled manner based on an inbuilt dataset. This is resistant to unfairness and disparity in all tests.

This step is significant as it eliminates the difference that may be caused by some factors like the quality of microphone or the background noise. It also enables the system to generate new inputs that do not necessarily come off training data and this helps to overcome the possibility of a model memorizing the patterns.

The other strength is that it facilitates stress testing. The models are all fed the same kind of generated audio, thus the difference in results is caused by the way the models behave and not the input.

C. Adversarial Proxy

The security of the system is its adversarial proxy. Its main purpose is with slightly altering the audio data prior to forwarding it to the models of classification. Such changes can be controlled and may also consist of well thought alterations to experiment with the system.

Understanding the performance of the models in both normal and challenging conditions can be checked by introducing this component between generation of data and processing of the models by this system. Such arrangement also ensures that all models undergo the same kind of changes thus ensuring that they are fairly compared in the strength they have.

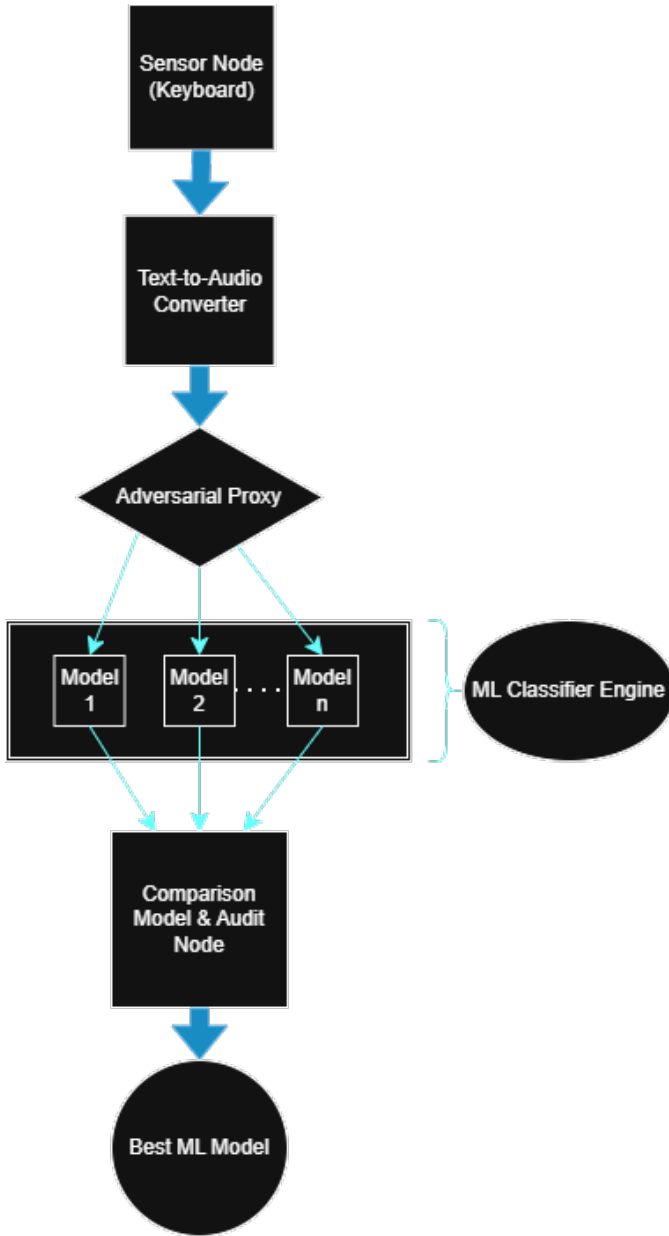


Fig. 1. System Architecture

D. ML Inference Engine

The Parallel ML Engine comprises a series of classification models that run on the load of the same input audio concurrently. Regardless of which model is used, every model does the same thing with the extracted features which can be either MFCC sequences or Log-Mel spectrograms and makes independent prediction as to the keystroke. Time and spectral patterns are learned by deep learning architectures like LSTM [14], CoAtNet [10],[17] and LogMel CNN [15],[16] over the input, whereas classical models like SVM and Random Forest directly classify structured representations of features. This parallel configuration allows comparing model performance directly in the same conditions and is the basis of the eval-

uation of robustness and further model selection in the audit layer.

E. Audit and Comparison Layer

The last stage of the system is the audit and comparison layer. It gathers and analyses the results of all the models. It does not compare each model independently but rather compares the predictions of the models to determine which model will be best in the different conditions, both normal and disturbed.

The system collects performance outcomes and predictions of all the models and compares them against each input. This can be used to determine which model is the most reliable or strong in such a situation. The analysis is not only restricted to the quality of the accuracy, but also the quality of the models to react to the disturbances or changes in the data.

After identifying the best-performing model and results, the data is processed into a hashed format and stored safely in a blockchain recorded ledger. This guarantees that the findings are immutable and may be always checked.

This layer does more than simple performance testing because it involves comparing and protecting storage. It introduces responsibility and traceability and ensures that the outcomes can be verified and recreated without the danger of being manipulated.

F. Processing Pipeline

1. The system has a very clear step by step procedure: 2. The input of keystroke is defined at the sensor node. 3. The TTA converter converts the input into audio. 4. Adversarial proxy can use some controlled modifications on the audio. 5. Several models are used to handle the modified input simultaneously. 6. The predictions are contrasted against the accurate results. 7. The findings will be documented to be analysed and compared.

The most successful model is found, and the performance information of the model is turned into a hash, and the information is stored safely behind a blockchain-based ledger.

This will make all models be tested under similar conditions. Consequently, any variations in performance arise not due to any differences in the data or processing steps but because of the models.

IV. METHODOLOGY AND EXPERIMENTAL DESIGN

The experimental framework will be used to determine the performance of the models and their health within the controlled and consistent conditions. Training and testing of all models are done on the same data pipeline thus any disparity in performance is not due to variations in data processing or the actual data itself but due to the model itself.

A. Feature Extraction

Samples are first put in an organized format and then fed to the models. There are two kinds of features that are employed:

- MFCC which provides a small overview of the sound.
- Log-Mel spectrograms, which indicate the changes of sound with time.

- Every audio sample has been made to fit to identical duration and format.
- This ensures that all the inputs are of the same type hence they can be easily processed by different models without any problem brought about by different sizes of audio.

B. Model Groups

The system consists of various categories of models to encompass most tactics:

TABLE I
MODEL GROUPING

Model Type	Description and Models
Conventional models	Support Vector Machines (SVM) [5],[6] and the Random Forest which operates on extracted features.
Deep learning models	LSTM [14] , Log-Mel CNN [15],[16] and Skypetype [7],[19] are the sequential and convolution-based deep learning models.
Hybrid models	CoAtNet-based models which combines and attention mechanisms (CoAtNet and CoAtNet Self-Attention). [10],[11],[17]

The training process of all models is based on the same dataset and feature extraction. This is to make sure that any variation in performance is attributed to the way the models learn and process the data but not the way the input is processed.

C. Single Keystroke Constraint

A critical limitation to the experiment is that deep learning models do not take a sequence of keystroke (entirely) as an input. Rather, individual samples of keystroke are used in all models. There are two strengths of this approach. First, it makes everything the same among all models, even those models that have an input size that is fixed. Second, it facilitates analysis at a more specific level, and it is easy to establish which keys are more inclined to be falsely classified in both normal and modified circumstances.

D. Experimental Setup

The procedure of all experiments is the same:

- A training dataset is used to train the models following the same processing steps.
- The monitoring of performance during training is implemented by validation data.
- The ultimate testing is done on a test set held out.
- The batch sizes, training time, and optimisation parameters are chosen in a way that allows the convergence with stable convergence without over-fitting. Although the model architectures in particular are varied, the training and evaluation procedure is the same process that can be applied to all experiments.

E. Adversarial Evaluation Setup

During the testing, controlled changes are introduced in order to test the robustness of the models. These transformations are made on the input features and then the models are applied. The analysis is conducted in two circumstances:

- Baseline condition: models are run on clean, unaltered inputs.
- Adversarial condition: The inputs are altered with controlled changes.

It is necessary to mention that not every model is compatible with any type of attacks. Gradient-based attacks can only be applied to the deep learning models and hybrid models due to the need of having models which can undergo differentiation. Other techniques, like in the frequency domain, are used to test traditional models, which do not use gradients.

This strategy will mean that every model is tested in a manner that is appropriate to its design but at the same time, they compare in a very fair and meaningful manner within the whole system.

F. Evaluation Approach

The experimental design is aimed at conforming both the accuracy of the models and the disturbance-conformability of the models. The system demonstrates the behavior of the various models under the effect of deliberate alteration of data by comparing clean and modified input results.

This is not only the method of considering accuracy. It also checks the stability and reliability of the models particularly in cases where the inputs are intentionally distorted.

V. THREAT MODEL AND ADVERSARIAL ATTACKS

A threat model that assumes an adversary with the capacity to modify input representations to influence a model prediction is used to evaluate the system. Instead of just utilising noise to disturb the system, the attacker aims to introduce organised perturbations that regularly result in misclassification while preserving the general properties of the input. The evaluation considers several levels of attacker capabilities, ranging from more constrained scenarios where the only thing that can be altered is the input representation to direct access to model gradients. This makes it possible for the system to capture both powerful and realistic attack scenarios.

VI. ADVERSARIAL ENVIRONMENT

A. Attacks Based on Gradients

Gradient-based techniques are used to generate adversarial perturbations for models that supports differentiation. By locating directions in the input space that maximise prediction error, these methods take advantage of the sensitivity of the model's decision boundary.[18]

- FGSM, or the Fast Gradient Sign Method: FGSM is a one-step attack that modifies the input in the direction of the gradient of the loss relative to the input. Because it exposes flaws in the model's decision surface with little

computational overhead, it offers a baseline measure of model vulnerability.

- The Basic Iterative Method, BIM: BIM extends FGSM by repeatedly applying the perturbation in smaller stages. This allows the attack to progressively enhance the input, leading to more successful misclassifications, while maintaining control over the perturbation's amplitude.
- Projected Gradient Descent (PGD): PGD is a multi-step attack that repeatedly alters the input inside a constrained region. After each phase, it projects that disturbed input back into a limited space to provide more reliable and robust adversarial samples. This method is considered a more dependable kind of iterative attack and is used to evaluate worst case model behaviour.

Only differentiable models, such as deep learning and hybrid architectures, can use these gradient-based techniques. These assaults don't work with models that don't have differentiable structures because they rely on gradient information.

B. Frequency Domain Attack

Each model responds to adversarial perturbations in a unique way. A key distinction arises when a model is differentiable: Differentiable models, such as deep learning and hybrid architectures, facilitate gradient-based attacks such as FGSM, BIM, and PGD. Since Support Vector Machines, Random Forest classifiers, and other non-differentiable models do not provide gradient information, these methods cannot be used to evaluate them.

For non-differentiable models, robustness is assessed using perturbations that are independent of gradients, such as frequency domain alterations. This ensures that every model is evaluated under conditions appropriate to its structure while maintaining consistency throughout the whole experimental framework.

C. The Attack Goal

The goal is the same for all attack techniques: to assess the ease with which controlled input manipulation can change model predictions. The system evaluates each model's accuracy, stability, and dependability by contrasting its performance in both clean and disturbed environments.

This method enables the analysis to identify variations in the models' reactions to adversarial pressure, revealing both their advantages and disadvantages in practical deployment situations.

VII. EVALUATION METRICS

The evaluation framework is designed in a way that can quantify the usual classification functionality the strength in specific adversarial circumstances. Instead of using one metric, there are several complementary measures which are employed to measure various aspects of model behaviour.

A. Baseline Performance Metrics

In clean experimental scenarios the models are evaluated based on conventional measures of classification.

Accuracy measures the ratio of correct keystrokes in the whole data set as this metric is a holistic measure of performance, but it does not measure the distribution of errors among the diverse classes.

Precision refers to the number of correct instances that are predicted to belong to a given set of predictions, and this is particularly relevant when dealing with keystroke classification with some keys having acoustic signature similarity and thus higher rates of confusion.

The F1-score combines precision and recall providing a harmonic mean thus providing a more reliable measure of performance in untypical situations with imbalance of classes or heterogeneous patterns of misclassification.

These metrics are compared to all variants of the model to set a reference point before the perturbation of adversarial examples is introduced.

B. Adversarial Evaluation Metrics

The system also evaluates model behaviour under adversarial conditions besides determining its behaviour at baseline performance.

The key measure of this, is the adversarial accuracy. It measures the percentage at which the adversarial perturbation can lead to an erroneous prediction of the model. High adversarial accuracy indicates that one is more vulnerable, and low values imply greater resistance to perturbation.

This metric therefore focuses the focus of performance at normal conditions to the vulnerability to manipulation. It provides a direct measure of robustness and is especially useful where it is used to compare models with similar baseline accuracy.

C. Per-Class Analysis

To further stretch the insights on the model behaviour, the performance is analysed in granular class level. This analysis aims at determining those keystrokes that are repeatedly misclassified and monitoring the dynamics of the misclassification patterns in adversarial conditions. Some of these keys can have similar acoustic characteristics, and thus, enhance their likelihood of confusion. Through analysis of per-class performance, it is possible to identify systematic weaknesses in the model instead of viewing all the errors as homogenous.

D. Comparative Evaluation

Every performance measure is tested in clean and hostile conditions, allowing a straightforward comparison of:

- Unperturbed performance and perturbed performance.
- Differences in different model architectures.
- Differences in strength in the various attack methodologies.

The framework can represent a more coherent model of model performance by incorporating both traditional performance

measures and adversarial ones. This method is such that conclusions are found not only by general accuracy, but also by the consistency by which every model works, when its operation is combined with controlled perturbation.

VIII. RESULTS

This section shows the performance of all the implemented models in clean and adversarial conditions. They are evaluated based on the classification accuracy and attack success rate (ASR), which makes it possible to directly compare the model robustness against various attack types.

A. Clean Performance

Classification report:					
	precision	recall	f1-score	support	
a	0.9828	0.9661	0.9744	59	
b	0.9355	0.9831	0.9587	59	
c	0.9828	0.9661	0.9744	59	
d	0.9667	0.9831	0.9748	59	
e	1.0000	0.9831	0.9915	59	
f	0.9821	0.9322	0.9565	59	
g	0.9492	0.9492	0.9492	59	
h	1.0000	0.9322	0.9649	59	
i	0.9833	0.9833	0.9833	60	
j	0.9500	0.9661	0.9580	59	
k	0.9344	0.9661	0.9500	59	
l	1.0000	1.0000	1.0000	59	
m	0.8529	0.9831	0.9134	59	
n	0.9219	1.0000	0.9593	59	
o	1.0000	0.9831	0.9915	59	
p	0.9667	0.9831	0.9748	59	
q	0.9831	0.9667	0.9748	60	
r	0.9219	1.0000	0.9593	59	
s	0.9167	0.9322	0.9244	59	
space	0.9833	1.0000	0.9916	59	
t	0.9344	0.9661	0.9500	59	
u	0.9643	0.9000	0.9310	60	
v	0.9796	0.8000	0.8807	60	
w	1.0000	0.9000	0.9474	60	
x	1.0000	0.9333	0.9655	60	
y	0.9508	0.9831	0.9667	59	
z	0.9219	0.9833	0.9516	60	

Fig. 2. Per-Class Baseline Results (CoAtNet Self-Attention)

Clean (unperturbed) test data was initially tested on all models in order to develop a baseline. The findings demonstrate that the deep learning models were always more accurate than the traditional machine learning models.

Of all the architecture types, CoAtNet architecture showed the best baseline performance, and it was the one that yielded the highest classification accuracy. LSTM, Skypetype and CoAtNet with self-attention were also close, which means that hybrid convolution-attention networks are effective when it comes to capturing the temporal and spectral attributes of audio signals representing keystroke.

In the LogMel CNN, there was a stable performance with the benefit on frequency-domain representations, however,

there was no higher sequence modelling capacity in the LSTM. On the contrary, classical models like SVM and Random Forest also attained lower baseline accuracy, as they only had limited capability in capturing the temporal dependence in MFCC sequences. Skypetype model [7],[19] also exhibited a similar pattern, doing well and not as good as the deep learning methods.

In general, the clean comparison shows that the models that can extract the temporal structure based on the sequential data have an obvious benefit in the process of keystroke classification.

B. Performance under Adversarial Attacks

Attack	Epsilon	Accuracy	Precision	F1-score	ASR
Clean	-	0.9181	0.9220	0.9179	0.0819
FGSM	0.005	0.9163	0.9205	0.9161	0.0837
BIM	0.005	0.9163	0.9205	0.9161	0.0837
PGD	0.005	0.9163	0.9205	0.9161	0.0837
FGSM	0.01	0.9150	0.9193	0.9148	0.0850
BIM	0.01	0.9150	0.9193	0.9148	0.0850
PGD	0.01	0.9150	0.9193	0.9148	0.0850
FGSM	0.02	0.9131	0.9177	0.9130	0.0869
BIM	0.02	0.9131	0.9177	0.9130	0.0869
PGD	0.02	0.9137	0.9182	0.9135	0.0863
FGSM	0.05	0.9062	0.9116	0.9061	0.0938
BIM	0.05	0.9038	0.9092	0.9037	0.0962
PGD	0.05	0.9038	0.9092	0.9037	0.0962

Fig. 3. Adversarial Results (LSTM)

The strength of each model was tested to withstand a variety of adversarial attacks, including FGSM, BIM, PGD as well as DFT-based perturbations. The findings have demonstrated distinct variations in the reaction from models to adversarial inputs.

Gradient-based attacks like FGSM, BIM and PGD, were effectively used in the case of deep learning models. The effect of these attacks was seen to have a range of degradation in accuracy depending on the strength of the attack and the model architecture. PGD had the best effect on a regular basis because it was an iterative method whereas FGSM caused a relatively smaller fall in performance.

This indicates that LSTM model was the most resistant to this kind of attacks, and it's accuracy varied comparatively smoothly with the level of perturbation. This implies that its ability to learn in a sequence helps in enhancing robustness to small perturbations in MFCC features.

The models of CoAtNet were of moderate strength. Their performance was good in clean conditions but when faced with more significant attacks like those of PGD their performance was affected in a more pronounced way. The addition of the self-attention only enhanced performance marginally without eliminating adversarial vulnerability completely.

The Skypetype Model on the other hand is of a similar strength as the CoAtNet models, however it's adversarial surface is restricted to running DFT attacks as it is of a base

of SVM (non-differentiable) and hence, gradient-based attacks aren't possible.

The LogMel CNN was sensitive to perturbation especially when it was attacked in an iterative manner. It is more vulnerable to adversarial noise to break frequency pattern as it relies on spatial feature extraction.

In the case of conventional machine learning models, gradient based attacks like FGSM, BIM and PGD were inapplicable since they do not have differentiable structures. Consequently, the models were tested with different mechanisms of attack, including DFT-based perturbations.

In this scenario, SVM and Random Forest exhibited noteworthy performance decreases, which reflect that they are not very robust to manipulating frequency-domain tests.

The Skypetype model had the same trend in that it had low performance when there was an attack situation and this indicates that it was sensitive to altered acoustic features.

C. Comparative Analysis Across Models

- A straightforward comparison between all models show a distinct trade off between baseline performance and robustness.
- The LSTM Model also had the best trade-off, indicating high levels of clean accuracy and a high level of adversarial attack resistance.
- CoAtNet and CoAtNet with self-attention provided a good baseline performance, but were more susceptible to more powerful attacks.
- The LogMel CNN was also reliable in clean conditions but was more sensitive to adversarial perturbation.
- The performance of the traditional models (SVM and Random Forest) and the Skypetype model was lower in terms of baseline performance, and had a lower ability to endure attack conditions.

In all models, PGD proved to be the best attack that at all times delivered the best attack success rates. FGSM was least effective with BIM offering middle behavior.

These findings affirm that the model architecture contributes greatly to defining robustness especially when working in contexts that involve adversarial manipulation of input features.

D. Best Model Selection and Audit Outcome

Rank	Model	CleanAcc	MeanAdvAcc	MeanASR	RobScore
1	LSTM	91.81%	91.38%	0.48%	91.33%BEST
2	CoAtNet	97.81%	89.54%	8.46%	89.35%
3	Skypetype	95.75%	81.20%	15.19%	80.56%
4	CoAtNet + Self-Attention	96.00%	71.17%	25.86%	70.14%
5	SVM	96.06%	58.50%	39.10%	56.96%
6	Random Forest	90.62%	57.30%	36.78%	53.84%
7	LogMel CNN	67.38%	48.17%	28.51%	38.87%

Fig. 4. Best Model Selection

After the assessment stage, the audit component compared the model outputs and the strongest performer was identified. The sampling was done through a mixture of clean accuracy, adversarial performance, and general robustness score. The LSTM model was found to be most relevant model since it

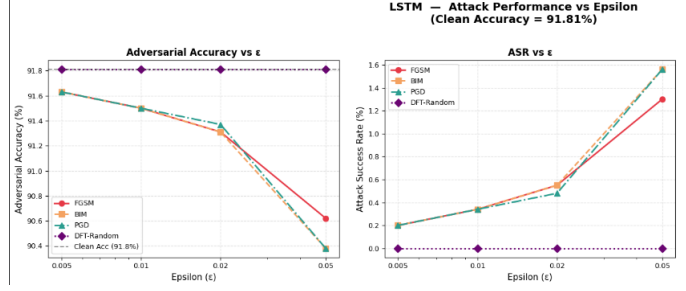


Fig. 5. Graphs - LSTM

had the highest score of the robustness and the most stable performance in all the test circumstances. It demonstrated high resistance to adversarial perturbation with great classification accuracy. The overall result of the selection was stored along-

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Best Model      : LSTM
Clean Accuracy  : 91.81%
Robustness Score : 91.33%
Mean ASR       : 0.48%
SHA-256 Hash   : e73ed8226776bc3a14746f57a866f1b9a04688507357bef9ebe305d447232120
Transaction Hash : 0xdfb410eb0a2b4c07d52ca768b94b0a6e2d1d50134dc8aa599abeb9736a3e5acd
Etherscan URL  : https://sepolia.etherscan.io/tx/0xdfb410eb0a2b4c07d52ca768b94b0a6e
Timestamp      : 2026-03-19 15:13:17 UTC
All results saved to /content/results/

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Fig. 6. Blockchain Secure Storage

side related metadata, such as performance indicators, time stamps, and cryptography hashes. The mechanism was built based on a blockchain ledger to ensure that this information was stored safely so that one cannot manipulate the results of the analysis and make it verifiable.

This is done to ensure that there is a transparency in the choice of model and has a strong audit trail of validating experimental results.

IX. DISCUSSION

The findings show a distinct difference in the variability of different model structures to clean and adversarial inputs. An intensive relationship exists between these differences and the processing of temporal and spectral information in each model using keystroke audio signals.

A. Impact of Model Architecture on Robustness

One of the most important findings of the results is that the models which can represent the temporal dependencies show a better resistance in adversarial conditions. The LSTM model was significantly better than all other models, not only with respect to baseline accuracy, but also with respect to performance under perturbations.

This effect may be explained by the fact that the LSTM is a time-sequential model, and therefore, it is able to learn temporal sequence patterns and not just use a given set of feature representations. Consequently, small changes that are added to the individual frames will have lesser effects on the entire prediction.

By contrast, the convolution-based models, included the LogMel CNN, are also very dependent on spatial feature

patterns. Although they can work well in clean environments, the models are more vulnerable to adversarial noise that distorts frequency-domain structures. This is why performance has been decreasing with the increase in the number of attacks.

Intermediate behaviour is exhibited by the CoAtNet models, which are convolution-based models with attention mechanisms. They do well when using clean data, but are not as robust in attack as the LSTM. Self-attention addition does not address the problem of adversarial sensitivity completely, yet it does help the model slightly.

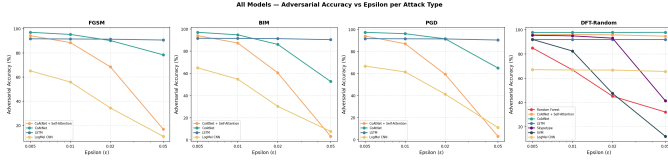


Fig. 7. Graph - All Models and Attacks

B. Behaviour of Gradient-Based Attacks

PGD generated the greatest model performance degradation of the applied attacks. It is far more effective than single step methods given it's iterative nature and thus able to optimize perturbations across multiple steps.

Single-step attack as FGSM only brought about minor perturbations and hence, it had reduced effect on accuracy. BIM, that employs numerous minor updates, demonstrated moderate performance, thus being between FGSM and PGD of the attack strength. This effectiveness gap in attacks shows the relevance of iterative optimization in the creation of adversarial examples. Models that seem to work well with FGSM can still have weakness when targeted by stronger and multi-step attacks like PGD.

C. Limitations of Traditional Machine Learning Models

Conventional models like SVM and Random Forest showed worse initial performance and also low robustness to attacks. The major weakness is that they cannot back gradient-based attacks since they lack differentiable structures. This also impacted derivative models like Skypetype (based on SVM).

Consequently, these models were tested by other perturbation techniques, including DFT based transformations. Whilst the performance is degraded, these attacks do not offer the same control and optimization as gradient-based techniques.

This brings out a more general weakness of the traditional models in adversarial research, they might seem more resistant to some forms of attack, but is more likely to be because of incompatibility than due to inherent resistance.

D. Role of Feature Representations

The MFCC features are very useful in model performance and vulnerability. MFCCs represent the valuable spectral features of the keystroke audio, and it makes them useful in the classification tasks. They however, also present structured feature space which can be exploited by adversarial attacks.

Minor changes of MFCC values may cause large changes in model predictions, especially those models, which are sensitive to local feature patterns. The latter is more noticeable in convolution-based models than sequence-based models like LSTM.

These findings indicate that the MFCC features can effectively be used in terms of classification, but they might need extra protective measures or data processing to enhance their risk-taking ability in antagonistic environments.

E. Effectiveness of Audit and Selection Mechanisms

Audit component is important in the interpretation of the model outputs and sufficient system behavior. The system can determine the model yielding the most overall balance of accuracy and robustness by comparing performance under a variety of conditions.

This objective is manifested in the choice of the LSTM model since it showed level performance in each case. A well-defined evaluation procedure is important to make sure that the choice of the model is not based on its ability to optimize the baseline accuracy, but also resilience to adversarial manipulation.

In addition, the secure storing of the outcomes with the help of a blockchain-based system helps increases transparency and confidence in the system. The performance data and hashes of associated performance results stored by the system makes sure that performance results cannot be changed after the evaluation, and a verifiable audit trail is available.

F. Key Insights

Based on the analysis conducted, a number of important insights can be made:

- Compared to purely convolutional model or the traditional model, the sequential models are robust in the keystroke classification.
- Single-step methods have a low level of effectiveness compared to iterative methods like PGD.
- Conventional machine learning models cannot work in adversarial environments because of their non-differentiable nature.
- Representation of features is very important in performance as well as vulnerability.
- Selection and audit process should be well organized to narrow down on the best model.

X. DESIGN JUSTIFICATION

The evaluation framework is intended to assess both conventional classification performance and robustness in hostile scenarios. Instead of depending on a single metric, other complementing metrics are used to include diverse aspects of model behaviour.

A. Security Considerations

The main goal of the system is to determine how vulnerable the models of the acoustic keystroke classification to input manipulation are. Adversarial testing is not seen as an additional post-processing step and is designed as a part of the pipeline.

The opportunity to use an adversarial proxy enables the controlled perturbations to be used consistently to all inputs. This will guarantee that models are tested by both a normal and a hostile environment, which is in a sense realistic because an attacker can seek to manipulate model predictions.

This is enhanced by the fact that a variety of attack methods has been used to support the evaluation. Gradient-based attacks reveal the vulnerabilities of differentiable models through their optimisation process, whereas frequency-domain perturbations identify the vulnerabilities that do not depend on the gradient information. The combination gives a more extensive perspective of possible attack surfaces.

Also, the audit layer provides the secure result handling mechanism. The system guarantees that the results of the evaluation will not be tampered with by ensuring that the result of the model comparisons is hashed and stored so that one can always verify the evaluation results independently. This provides an extra degree of accountability which is normally lacking in the normal model evaluation pipelines.

B. Resilience and Robustness

The concept of resilience is resolved through assessing model behaviour during controlled perturbations, and not the use of clean test data only. The system determines the extent to which the performance suffers due to changes in inputs thereby giving an insight on the stability of the different models.

One of the main design decisions includes having several model architectures that are varied in terms of being traditional classifiers, deep learning, and related attention-based models. This enables the system to make comparisons in the way in which various learning mechanisms react to adversarial pressure. There are cases where some of the models can be resistant to small perturbations whereas others can be highly degraded, despite a high base accuracy.

The fact that most of them use single keystroke inputs further corroborates this analysis. The isolation of each classification instance is possible to provide the system with the points of failure that are more likely to be identified and the keys or patterns where misclassification with the system is more likely to occur. The granularity level allows the weaknesses that may be masked in the sequence-level assessment to be identified.

Resilience is thus considered not just based on the general performance measures, but in terms of the dependability and similarity of predictions in diverse situations.

C. Innovation in System Design

The system provides a procedural method of assessing the models of acoustic keystroke classification through the integration of data generation, adversarial testing, and multi-model comparison in one pipeline.

A text-to-audio conversion step eliminates variability and ensures consistency of input data as it is a controlled way to produce input data. This enables the assessment to deal with model behaviour in lieu of exogenous variables like recording conditions.

The fact that heterogeneous types of models are included in the same framework allows directly comparing fundamentally different approaches such as traditional machine learning, deep learning, and hybrid architectures. This increases the area of analysis and does not result in conclusions which are restricted to just one type of model.

The other feature of the design is the direct management of model-specific constraints. Instead of applying the same method of evaluation to all the models, the system uses the attacks that fit each structural model. This preserves methodological validity but retains significant comparison within the system.

Lastly, there is the inclusion of secure result storage, which adds a further point of reliability. The system is taken a step further by integrating performance analysis with record keeping practices that are verifiable, and this is a more accurate representation of how these models would be applied in the real world.

D. Summary

There is a way of balancing performance testing and security testing by integrating adversarial testing at all levels of the pipeline. The system offers an encompassing system to evaluate accuracy and robustness by integrating the control input generation, variety of model architectures, and organized attack techniques. This method will make sure that the evaluation does not consider the effectiveness of the models only, but also the way they respond when subjected to conditions that put their stability to test.

XI. CONCLUSION

This project effectively develops and tests a complete end-to-end pipeline for the classification of keystrokes from an acoustic point of view, and focuses specifically on model robustness in the presence of adversarial perturbations, another facet of an understudied aspect of audio-based behavioural biometrics.

Extremely novel exquisitely diverse intuitions across large ranges of architecture in experiment reveal pretty apparent performance hierarchies. The model based on LSTs is proven to be the best and shows a consistent high classification accuracy on benign data and an outstanding resilience to a range of adversarial attacks. In contrast, transformer-inspired (CoAtNet) and convolutional (LogMel CNN) architectures, while being competitive under clean conditions, exhibit severe degradation under intensified perturbations. Traditional classifiers (SVM and Random Forest) are poor in evaluating its efficiency in terms of baseline performance and tolerance to adversaries and highlights the benefits of recurrent architectures in this setting for sequential acoustic patterns.

The systematic consideration of adversarial evaluation (especially via potent iterative approach methods like Projected Gradient Descent (PGD)) provides the sensitive identification of realities which are not captured by default accuracy metrics on clean data sets. These findings reinforce one of the essence: subject to security to mrss security program: powerful

model choice in milli security security types to rsen need affiliates appellate more than nominal performance in accounts for manipulability marae in reality.

A great contribution of this work is the novel integration of an audit and model selection, which African researchers beyond traditional and accuracy-driven pipelines. By comfortably quantifying performance in benign and adversarial regimes within the combined framework, the system self-governs the best model with the least.

By quantifying performance as hell and the world in the same framework, performance quantifies itself and accesses the least dependent model, as per the system. Crucially, the selection decision is immutably recorded via a blockchain-based mechanism, which gives them the tamper-resistant properties of transparency, verifiability and auditiability; attributes which are becoming increasingly crucial for trustworthy implementation of machine learning algorithms in authenticate systems.

Ultimately, this project is an advance in the field as it shows that secure adversarial testing and secure and auditable model governance can truly make for vastly more reliable solutions to the classification of keystrokes. The proposed framework not only addresses the risks inherent in the acoustic side channel attack it reduces the risk of such attacks, but it also creates a recipe book for the creation of resilient and accountable ML systems in privacy-critical and security-oriented settings. Future extensions might learn more about hybrid defenses (adversarial training/certified robustness), multimodal fusion and feasibility for real-time deployment to additionally impregnate such systems against evolving threats.

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XII. APPENDIX

A. Group Contribution

Every Member showed equal contribution in the work, with all the models and the attacks, and the blockchain integration responsibilities being split and shared so that we all learn together.

- Abdul Hamdhan Ahamed - (100/6)%
- Karthik S - (100/6)%
- Karuppasamy Karuppasamy - (100/6)%
- Nadeem Mohamed Abdul - (100/6)%
- Manav Sharma - (100/6)%
- Miran Mansuri - (100/6)%